# American Computer Science League

2022-2023 ● Contest 1: Short Problems Solutions ● Classroom Division

**1. Computer Number Systems**

$2023_{10}$ = 3 * 512 + 7 * 64 + 4 * 8 + 7
= 3 * 8^3 + 7 * 8^2 + 4 * 8^1 + 7 * 8^0
= $3747_8$

B. 3747

**2. Computer Number Systems**

$657383_{16} - 654321_{16} = 3062$

$3062_{16} = 11000001100010_2 = 11\ 000\ 001\ 100\ 010_2 = 30142_8$

D. 30142

3. **Recursive Functions**

$f(18) = f(18-3)+1 = f(15)+1 = 21+1 = 22$
$f(15) = f(15-3)+1 = f(12)+1 = 20+1 = 21$
$f(12) = f(12-3)+1 = f(9)+1 = 19+1 = 20$
$f(9) = f(9-2)-3 = f(7)-3 = 22-3 = 19$
$f(7) = f(7-2)-3 = f(5)-3 = 25-3 = 22$
$f(5) = 25$ Now substitute backwards.

E. 22

4. **Recursive Functions**

$f(12,21) = f(21-1,12+2)+1 = f(20,14)+1 = 12+1 = 13$
$f(20,14) = f(20-1,14+2)-3 = f(19,16)-3 = 15-3 = 12$
$f(19,16) = f(19-1,16+2)-3 = f(18,18)-3 = 18-3 = 15$
$f(18,18) = 18$ Now substitute backwards.

A. 13

5. **What Does This Program Do? (Branching)**

Since 4, 5, 7 is not a Pythagorean triple, the else is invoked and the area of the triangle is calculated by Heron's formula. The semiperimeter, s, is (4+5+7) / 2 = 8. Area = sqrt(s * (s − a) * (s − b) * (s − c))
= sqrt(8*(8−4)*(8−5)*8−7))
= sqrt(8*4*3*1)
= sqrt(96)
So ans = int(sqrt(96)) = 9.

B. 9

**6. Computer Number Systems**

| Decimal | Binary |
|---|---|
| 8 | 1000 |
| 9 | 1001 |
| 10 | 1010 |
| 11 | 1011 |
| 12 | 1100 |

| Decimal | Binary |
|---|---|
| 13 | 1101 |
| 14 | 1110 |
| 15 | 1111 |
| 16 | 10000 |
| | |

You get a total of 21 when you count all of the 1s from 8 to 16, inclusive.

B. 21

**7. Computer Number Systems**

$78B_{16} = 11110001011_2 = 3613_8$,
$ACE_{16} = 101011001110_2 = 5316_8$,
$F4A_{16} = 111101001010_2 = 7512_8$,
$2EF_{16} = 1011101111_2 = 1357_8$,
$BAD_{16} = 101110101101_2 = 5655_8$

The most odd digits is 4 in $1357_8$ which is $2EF_{16}$.

D. 2EF

**8. Recursive Functions**

$f(1) = 2$
$f(2) = 4$
$f(3) = 2 * f(2)/f(1) = 2 * 4/2 = 4$
$f(4) = 2 * f(3)/f(2) = 2 * 4/4 = 2$
$f(5) = 2 * f(4)/f(3) = 2 * 2/4 = 1$
$f(6) = 2 * f(5)/f(4) = 2 * 1/2 = 1$
$f(7) = 2 * f(6)/f(5) = 2 * 1/1 = 2$
$f(8) = 2 * f(7)/f(6) = 2 * 2/1 = 4$

So the sequence of numbers produced is: 2 4 4 2 1 1 2 4 4 2 1 1 …
This repeats in a cycle of 6 so f(25) = 2.

A. 2

**9. Recursive Functions**

$f(9) = f(9 + 1) - 1 = f(10) - 1 = 7 - 1 = 6$
$f(10) = 10 - 3 = 7$
$f(6) = f(6 + 1) - 1 = f(7) - 1 = 4 - 1 = 3$
$f(7) = 7 - 3 = 4$
$f(3) = f(4) - 1 = 0$
$f(4) = f(2) + 2 = -1 + 2 = 1$
$f(2) = 2 - 3 = -1$

So $f(f(f(9))) = f(f(6))$
$= f(3)$
$= 0$

E. 0

**10. What Does This Program Do? (Branching)**

A) The values 3, 6, 9 don't form a triangle.
B) The values 4, 5, 6 form an ACUTE and SCALENE triangle.
D) The values 6, 6, 6 form an EQUILATERAL triangle (as well as ISOSCELES) which are always ACUTE.
E) Though the other 2 both form an OBTUSE triangle, the values 6, 6, 10 (E) form an ISOSCELES triangle.
Only a=5, b=6, and c=10 form an OBTUSE and SCALENE triangle.

C. a=5, b=6, c=10